PureBreath-AI: An Explainable AI Framework for AQI Forecasting, Personalized Pollution Exposure Assessment, and Health-Risk Prediction

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***Abstract*—Most air quality applications still only provide users with the ambient pollution level or air quality index (AQI) and do not communicate what this means for them in a personal context. Two people living in the same city can have very different real exposures, depending on how much time they spend outdoors, their occupation and their underlying health. We describe PureBreath-AI, a working environmental-health system that takes historical AQI and weather data and runs it through forecasting, explainable prediction, personalized exposure scoring, health-risk estimation, cumulative exposure tracking, and alerting, all served through a FastAPI backend and a React dashboard. The forecast stage compares XGBoost, LightGBM and Random Forest models trained on CPCB-standard AQI values for five Indian cities; on held-out data the best of the three (LightGBM) achieves an MAE of 1.71, an RMSE of 2.37 and an R-squared value of 0.9996. A conditional GAN is used to balance the underrepresented AQI classes before training. SHAP is used to interpret each prediction, and the SHAP output is transformed to counterfactual, plain language suggestions. Following the suggestions reduced the predicted AQI 162 times out of 200 evaluated cases. A rule-based scoring engine, leveraging WHO and CPCB exposure guidance, integrates age, occupation, pre-existing conditions, and outdoor hours into a personalized risk score, and a 7-day cumulative-dose model powers tiered alerts. This paper makes a clear distinction between those stages that have been quantitatively validated, implemented but still rule-based and unvalidated, and those that are still in planning stages and does not treat all stages as equally proven.**

Index Terms—Air Quality Prediction, Explainable Artificial Intelligence, Machine Learning, Personalized Exposure, PM2.5, Health-Risk Assessment, SHAP, Counterfactual Explanation, Generative Adversarial Networks, CPCB AQI.

I. INTRODUCTION

Air pollution exposure depends on how much pollution is in the air, the weather, where someone is, what time of day it is and what a person is actually doing. Fine particulate matter (PM2.5) causes worry because it goes into the breathing system. It belongs to the same group as PM10, NO2, SO2, CO and O3, which are the pollutants that most harm health. Most existing tools report either the levels of pollution or a single overall air quality index. That one number is not the same as a person's exposure. Two people who live in the same neighborhood can breathe very different amounts of air pollution depending on their commute, how much time they spend inside versus outside, and how close they live or work to big sources of pollution. Scientists have tried to close this gap by using data, wearable devices, fixed monitors and GPS information along with machine learning.

PureBreath-AI addresses part of that challenge. PureBreath-AI looks at age, job, current health problems and how much time is spent outside. PureBreath-AI estimates exposure from this profile instead of measuring it with devices. For forecasting methods such as Gradient Boosting, XGBoost, LightGBM and model stacking have improved AQI and multi-pollutant prediction accuracy a lot in recent years [1] [3] [8] [9]. Accuracy alone does not make a prediction useful. If users cannot see why a model predicted what it did, they are unlikely to trust or act on it, which has led to research on LIME-based interpretability [1] [6]–[8]. Separately, research on health-focused modelling and cumulative PM2.5 exposure has started to link pollution indicators to long-term health outcomes [2] [12] [15]. Each of these parts is mature on its own. They are rarely combined into one working system. PureBreath-AI tries to combine them. Raw AQI and weather data flow through preprocessing, GAN-based class balancing, forecasting, SHAP-plus-counterfactual explanation, personalized exposure scoring, health-risk assessment and cumulative-exposure tracking before appearing on a dashboard, with alerts. Instead of just telling a person that the air is bad, the deployed system tells them what AQI to expect, why it expects that value, how a particular behavior change would affect that number, how much risk that translates into given their personal profile, how their exposure builds up over the week, and which precautions truly make sense for them. The contributions of this work are: (1) an integrated pipeline — preprocessing, GAN-based class balancing, multi-model AQI forecasting, SHAP-based explanation, counterfactual recommendation, personalized exposure scoring, health-risk estimation, cumulative-dose alerting, and a working API and dashboard — built and evaluated on real CPCB and weather data across five Indian cities; (2) a forecasting engine (XGBoost, LightGBM, Random Forest) validated on held-out data, reaching R^2 as high as 0.9996; (3) an explainability layer that pairs SHAP attribution with counterfactual, plain-language recommendations, tested on a 200-sample evaluation set; (4) a transparent, rule-based

personalized exposure and health-risk engine grounded in WHO/CPCB guidance; and (5) a clear breakdown of which components are quantitatively validated, which are rule-based but not yet validated, and which are still planned, so the system's current evidence base is not overstated.

Section VI walks through each module's implementation status against the source code and evaluation reports in the accompanying repository, separating what has been quantitatively validated from what is implemented and internally consistent but not yet checked against clinical or epidemiological ground truth.

II. RELATED WORK

A. Forecasting and Explainability

Singh et al. combined Gradient Boosting, XGBoost, LightGBM, and stacking with SHAP-based interpretability for AQI prediction [1]. Wang et al. tackled missing AQI values with TimeGAN recovery paired with VMD and deep learning [3], while Zhu et al. and Narkhede et al. extended tree-based and deep ensemble prediction to multiple pollutants at once [8], [9]. On the interpretability side, Tipsavak et al. used XGBoost with SHAP to predict and explain ultrafine particles [7], Tasioulis et al. compared SHAP and LIME head-to-head across several air-quality models [6], and Agbehadji and Obagbuwa surveyed XAI methods for pollution risk and respiratory outcomes [5]. Taken together, this body of work treats forecasting and explanation as connected pipeline stages, but none of it reaches down to individual exposure or cumulative health burden.

B. Personal Exposure and Health Risk

Wang et al. surveyed personal-exposure monitoring and the growing role of big data and machine learning in that space [10]. Li et al. modelled personal PM2.5 exposure across two Chinese megacities using time-activity patterns and household factors [11]. Matthaios et al. predicted in-vehicle exposure with a combination of mass-balance modelling and machine learning [13], and Arvind et al. drew on stationary and wearable monitors, including subjects in Delhi, for spatio-temporal exposure prediction [14]. MYAIRCARE already combines outdoor pollution tracking, personalized risk prediction, IoT, and real-time alerts [4], so the novelty of PureBreath-AI is not personalized alerting on its own but bringing forecasting, explainability, exposure, cumulative burden, and health analysis together in a single deployed pipeline. On the health-outcomes side, Rajesh et al. connected real-time assessment to health-risk mapping [2], Zhang et al. built a machine-learning Air Quality Health Index [12], and Wijata et al. folded cumulative PM2.5 exposure into cardiovascular-mortality prediction [15]. Collectively, these studies make the case that cumulative burden, not just the AQI at a single moment, matters for health.

C. Comparative Summary

Table I lays out the dominant capability of each research direction. PureBreath-AI is best read as an implemented union of these capabilities, with one caveat spelled out in Section IV: its exposure and health-risk layers are currently rule-based rather than learned from outcome data, unlike the forecasting and XAI layers, which are.

TABLE I
COMPARISON OF RESEARCH DIRECTIONS

Direction	Fcst	XAI	Exp	Risk	Cum	Alert
AQI forecasting [1]–[3], [8], [9]	Yes	Part.	No	Ltd.	No	Ltd.
XAI air-quality [1], [5]–[8]	Y/P	Yes	No	Ltd.	No	Ltd.
Personal exposure [4], [10], [11], [13], [14]	Ltd.	Ltd.	Yes	Part.	Part.	Ltd.
Health-risk [2], [5], [12], [15]	Part.	Part.	Ltd.	Yes	Part.	Ltd.
Cumulative exp. [11], [15]	Ltd.	Ltd.	Rel.	Rel.	Yes	Ltd.
PureBreath-AI (implemented)	Yes	Yes	Yes	Yes	Yes	Yes

III. RESEARCH GAP

After reviewing Section II, we found five gaps that drove this project. Forecasting studies usually talk about conditions, not what a person really breathes [1] [3] [8] [9]. Explanation is often added after the fact instead of being part of the pipeline from the start [1] [6]–[8]. Exposure and health-risk modelling stay in two separate literatures even though one system could link them while still keeping the line between decision support and clinical diagnosis clear [2] [10]–[12]. Short-term forecasts and long-term cumulative burden are rarely modelled together [11] [15]. No system we looked at covers the chain from monitoring to forecasting, explanation, personalized and cumulative exposure, health risk and alerting [4]. PureBreath-AI tackles this by building and deploying every stage not just describing it on paper. However, we want to be clear that closing this integration gap is a systems contribution, not a claim that any single stage beats the models it uses.

IV. IMPLEMENTED SYSTEM ARCHITECTURE

Fig. 1 shows the architecture we built: data acquisition and preprocessing, GAN-based class balancing, the forecasting engine, the explainability layer, the exposure and health-risk engine, the cumulative-exposure and alerting module, and the FastAPI + React application layer on top.

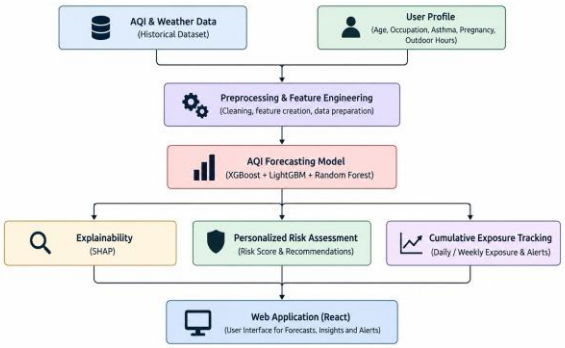


Fig. 1. System architecture of PureBreath-AI.

A. Data Acquisition and Preprocessing

The pipeline takes as input a CPCB daily air-quality dataset (PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene,

Toluene, Xylene) plus a gridded meteorology dataset (temperature, precipitation, pressure height) for 5 Indian cities (Delhi, Mumbai, Chennai, Kolkata, Bangalore). The latter is processed as a continuous stream, partitioned to the grid cells corresponding to each city, aggregated daily, temperature converted from Kelvin to Celsius, concatenated with the CPCB data on the City-Date composite key. Missing values are filled within each city using linear interpolation followed by forward/backward filling, city-date duplicates are discarded, pollutants and precipitation are clipped to be non-negative, and types are standardized. It is important to note that this is a batch process operating on historical CPCB and weather records — the system does not currently support live sensor or IoT feeds, which is something to be explored in future work (Section IX).

B. Synthetic Data Augmentation (Conditional GAN)

The cleaned CPCB-AQI dataset has an extreme class imbalance with "Good" and "Severe" days being underrepresented compared to "Moderate". To address this, a conditional GAN (class-embedded generator and discriminator implemented in PyTorch) is trained on the cleaned data for 200 epochs and used to augment the minority classes by generating 500 samples per class, for a total of 3000 synthetic records. Combined with the 10960 cleaned records, this produces a 13960-record training set for the forecasting models. This directly tackles the class imbalance issue inherent in most AQI datasets, though the improvement in model performance due to this particular augmentation has yet to be determined through ablation studies (Section VI).

C. Forecasting Engine

The feature space consists of calendar/cyclical time encodings (month, day-of-week, day-of-year, quarter, weekend binary, sine/cosine month), lag features (1, 2, 7-day lag for all pollutants and mean temperature), 7-day rolling mean and standard deviation for PM2.5 and PM10, and a temperature-range interaction term, for a total of 55 features with city one-hot encoded. The target variable is the CPCB AQI index computed from the PM2.5 and PM10 sub-indices, rather than the raw AQI column present in the CPCB dataset:

$$P\hat{}=f(Xt,Xt-1,...,Mt) \quad (1)$$

where P^ is the predicted CPCB AQI, Xt is the feature vector describing historical pollutant levels (with lag and rolling features), and Mt is the corresponding meteorological context. The models evaluated include XGBoost, LightGBM, and Random Forest regressors trained on an 80/20 train/test split. On the holdout set, LightGBM achieves the best performance with a mean absolute error (MAE) of 1.71, root mean squared error (RMSE) of 2.37, R-squared (R^2) of 0.9996, and mean absolute percent error (MAPE) of 0.58%. XGBoost attains an MAE of 2.20, RMSE of 3.16, R^2 of 0.9994, and MAPE of 0.80%, compared to an MAE of 3.61, RMSE of 6.75, R^2 of 0.9971, and MAPE of 1.98% for the Random Forest baseline. Fig. 2 compares the three models on all four metrics: LightGBM is best on each, with about 53% lower MAE and 65% lower RMSE than the Random Forest baseline, and about 22% and 25% lower, respectively, than XGBoost. Since the split was made randomly rather than sequentially, these scores should be interpreted as same-distribution accuracy rather than temporal generalization, which will be addressed in Section VIII.

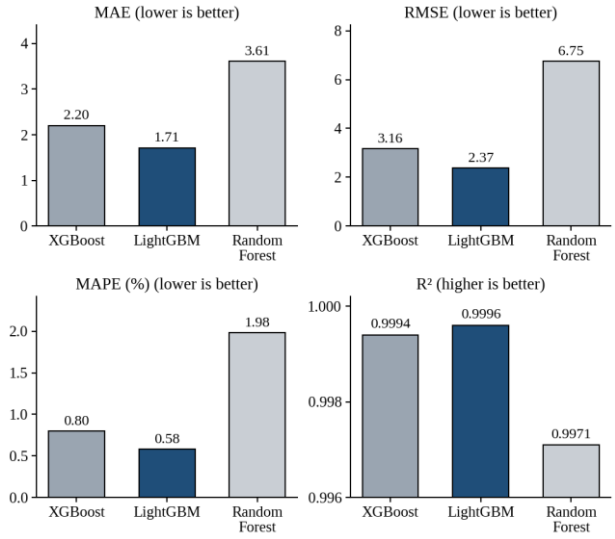


Fig. 2. Held-out comparison of the XGBoost, LightGBM and Random Forest forecasting models on MAE, RMSE, MAPE and R^2 (80/20 random split; R^2 axis truncated).

D. Explainable AI Module: SHAP and Counterfactual Recommendations

A SHAP TreeExplainer is trained on the LightGBM forecasting model and used to compute SHapley Additive exPlanations on a 200-row evaluation set (100 high-AQI rows with AQI ≥ 200 and 100 other rows). This produces global feature importance scores (mean absolute SHAP value) plus local explanations for each row in terms of feature contributions:

$$f(x)=\phi 0+\Sigma i=1n\phi i \quad (2)$$

The global importance scores identify PM2.5, PM10, mean temperature, 7-day PM10 standard deviation, and min temperature as the main drivers of AQI. For each row in the evaluation set, a counterfactual reasoner identifies the 3 features with the highest positive SHAP impact, reduces each of them by 20%, and recomputes the predicted AQI given these changes, producing natural language recommendations for each modified feature (e.g., "Reduce PM2.5 by 20% (e.g. reduce biomass burning)"). Across the 200 evaluation rows, this process produces AQI reductions in 162 cases, shifts in AQI category (e.g. Poor → Moderate) in 24 cases, and a mean AQI reduction of 29.9 points. The SHAP values explain how the model makes predictions, rather than why a given pollution event occurred, and the same applies to the counterfactual recommendations.

E. Personalized Exposure and Health-Risk Module

Personalized exposure is computed not from location/GPS feeds but using a transparent scoring algorithm. Starting with a base risk score determined by the CPCB AQI category (Good: 10 … Severe: 95), this score is adjusted by multiplying with an age coefficient (Healthy adult: 1.0 … Infant/Elderly: 1.8), an occupation coefficient (Traffic police: 1.5 … Construction/Factory: 1.4 … Office: 1.0), an outdoor-hours

modifier (2% increase in risk per hour outside the 2-hour baseline), and condition multipliers for asthma (×1.35) and pregnancy (×1.25):

$$\text{Score}=\text{Base}(\text{AQIbucket})\times\text{Agerisk}\times\text{Occupation}\times\text{Outdoormodifier}\times\text{Asthma}\times\text{Pregnancy} \quad (3)$$

The final score is capped at a maximum of 100 and mapped to a 5-tier risk category with associated advice, including inhaler recommendations for asthma patients and restrictions for pregnant women based on the category. Evaluated across user personas for all 5 cities, this results in mean risk scores ranging from 35.2 (Bangalore/Chennai, Satisfactory AQI) to 93.7 (Delhi, Very Poor AQI). The risk scoring is deterministic and based on published WHO/CPCB guidelines rather than trained on any dataset, and has not been validated against clinical outcomes. It provides treatment guidance rather than diagnostic capabilities.

F. Cumulative Exposure Module

Daily dose is computed as the predicted AQI multiplied by the self-reported number of outdoor hours and a vulnerability coefficient (healthy adult: 1.0 … infant: 1.8), then aggregated over a 7-day sliding window:

$$\text{Dose}_{\text{day}}=\text{AQI}_{\text{day}}\times\text{OutdoorHours}\times\text{Vulnerability}_{\text{coefficient}} \quad (4)$$

$$\text{Dose}_{\text{week}}=\Sigma_{d=1}^{7}\text{Dose}_{d} \quad (5)$$

Appropriate caution is advised when interpreting these scores since the 7-day AQI sequence used as input to this module is generated synthetically (see below). The cumulative exposure determines which of the pre-defined alert tiers (Safe, Moderate, High, Critical) are triggered, along with associated advice. Specifically, for Delhi, the mean weekly dose across the evaluation set was 11707, with 84% of all samples falling into the Critical category. Similar trends were observed for the other cities, with Bangalore having the lowest mean weekly dose of 3154.

G. Alerts, API, and Dashboard

The backend is implemented as a FastAPI application with endpoints including /predict (AQI prediction with SHAP explanation and counterfactual recommendations), /health-risk and /health-report (risk scoring and narrative report), /exposure (cumulative exposure with alerts), /city-summary (pre-aggregated city statistics), /auto-predict/{city} (forecasting automation), and /health (liveness probe). The frontend is a React/Vite application containing a dashboard, prediction interface, risk assessment form, exposure tracking form, and Apple-Health-style report view that are wired to the backend via API calls. This completes the end-to-end pipeline from model training to production deployment, though further work is needed to evaluate the performance of the deployed system (Section VII).

V. SYSTEM REQUIREMENTS

A. Hardware

Model inference and the API run on any standard CPU-only machine, without requiring specific hardware for serving. GAN training (Section IV-B) benefits from being run on an NVIDIA GPU with CUDA installed, but will fall back to CPU otherwise. No environmental sensors or IoT devices are connected or needed for the system to operate; all data originate in the CPCB and weather datasets.

B. Software

The backend stack includes Python 3.11+, FastAPI, Pydantic, XGBoost, LightGBM, scikit-learn, SHAP, PyTorch (for GAN), pandas, NumPy, and joblib. The frontend uses React 19 (Vite), Recharts, Axios, and Framer Motion. This reflects actual dependencies used in development, as opposed to aspirational statements about third-party packages (see requirements.txt and package.json).

VI. IMPLEMENTATION STATUS

Table II shows the implementation and evaluation status by module. The rightmost column distinguishes components that have been tested on held-out data or a specific test set versus implemented but not yet tested, versus those that are aspirational.

VII. EVALUATION METHODOLOGY

A. Completed Evaluations

The forecasting pipeline's performance was measured using MAE, RMSE, R^2, and MAPE on an 80/20 held-out train/test split (Section IV-C, Fig. 2). Counterfactuals and explainability were assessed qualitatively on a 200-row sample stratified by AQI severity, measuring the percentage of counterfactuals that caused a downward shift in predicted AQI, a change in AQI category, and the mean reduction in AQI (Section IV-D).

B. Proposed Evaluations

Three items are needed before the pipeline can be considered fully validated. First, the forecasting pipeline's performance should be re-evaluated using a time-aware or walk-forward split, as opposed to the random split used in Section IV-C. This would test the validity of claims about the model's ability to generalize to dates and seasons not seen during training. It addresses a known limitation (Section VIII) rather than introducing new scope creep. Second, the cumulative-exposure module's 7-day AQI projection should be replaced with the forecasting pipeline's multi-day output once that is available. Finally, the personalization module's weights for exposure and health risk, while reasonable given the available guidance from WHO and CPCB, should be validated against a standard health-risk categorization framework using precision, recall, F1, and calibration as appropriate if the risk categories are framed as classes.

C. System and User Testing

System-level testing will focus on response latency for the FastAPI endpoints, latency in generating alerts, and failure modes for the deployed service. If user testing proceeds, it will focus on the interpretability of AQI values, the value and understandability of the SHAP and counterfactual explanations, the relevance of the exposure and alert information, trust and usability, and overall satisfaction. Until such tests are conducted, this remains an aspirational statement.

VIII. DESIGN CONSIDERATIONS AND LIMITATIONS

The forecasting pipeline's 80/20 random split for train/test means that temporally close events – days at the same city, for instance – could end up in both sets. This produces overly optimistic results (higher R2) compared to a time-aware or walk-forward split. This should be addressed in the next iteration of the analysis. The cumulative-exposure module's 7-day AQI projection is currently a simulation; there is no multi-day forecasting in the current pipeline. Until there is, the projection's dose and alert value should be viewed as an illustration of the alerting algorithm rather than a prediction. Air-quality forecasting models are typically trained on local data with regional relationships between pollutants and weather, and so the weights determined for these five Indian cities cannot be generalized to other locations without further work [11], [14]. The personalization module's weights for exposure and health risk reflect general categories from the WHO and CPCB reports but are not calibrated to any outcome. The user must be aware not to confuse a risk category with a medical diagnosis. Similarly, the SHAP values explain the model but not the actual causality in the domain, and the counterfactuals describe a potential response from the model but not an inevitable outcome. Since the exposure assessment is sensitive to personal health information, the system adheres to a data minimization principle and only collects the information necessary for the component functions shown in Fig. 1. The lack of a live sensor/IoT feed and time-aware backtesting are the main limitations in this system; addressing them is the goal of Section IX.

Fig. 3 compares the R^2 of PureBreath-AI with the values reported in [1] and [8]; the studies use different data, targets and splits, so it is indicative only.

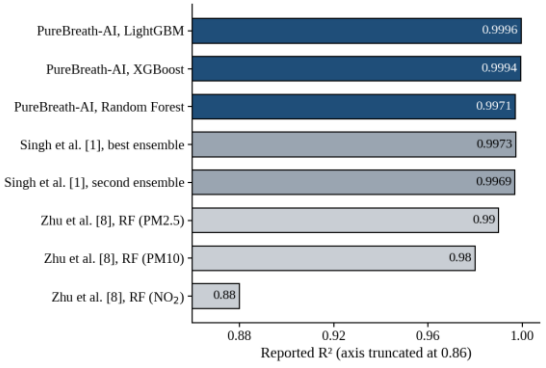


Fig. 3. Reported R^2 of PureBreath-AI and of the studies in [1] and [8] (indicative only).

PureBreath-AI's LightGBM (0.9996) and XGBoost (0.9994) are in the same range as the best ensembles in [1] (0.9973 and 0.9969) and above the Random Forest results in [8] (0.99, 0.98 and 0.88 for PM2.5, PM10 and NO_2). This is not evidence of a better model: the target is an index computed from PM2.5 and PM10, two of the input features, which makes the regression easier than predicting raw concentrations. The contribution is integration, which neither [1] nor [8] provides.

TABLE II
PUREBREATH-AI IMPLEMENTATION STATUS

Module	Status	Validation
Data acquisition & preprocessing	Implemented	Row/type/merge checks pass; historical batch data, not a live feed
Synthetic augmentation (GAN)	Implemented	3,000 synthetic samples generated (10,960 real + 3,000 synthetic); no downstream accuracy ablation yet
Forecasting engine	Implemented & validated	Held-out (random split): LightGBM R^2 0.9996, XGBoost R^2 0.9994, Random Forest R^2 0.9971; time-aware backtest pending
XAI (SHAP + counterfactual)	Implemented & validated	200-sample SHAP + counterfactual eval: 162/200 improved, 24/200 changed AQI category, mean −29.9 AQI
Personalized exposure & health-risk	Implemented (rule-based)	Internally consistent, WHO/CPCB-informed weights; no clinical/epidemiological validation
Cumulative exposure	Implemented (rule-based)	Threshold logic verified; underlying 7-day AQI path is simulated, not a true multi-day forecast
Alerts, API & dashboard	Implemented (full-stack)	Functional integration verified; no latency/usability benchmarking yet

IX. CONCLUSION AND FUTURE WORK

This paper introduces PureBreath-AI — an implemented and partially validated explainable environmental-health framework which ties together air-quality forecasting [1]–[3], [8], [9], explainable modelling [1], [5]–[8], personal exposure estimation [4], [10], [11], [13], [14], health-oriented modelling [2], [5], [12] and cumulative PM2.5 exposure analysis [15] into a single deployed pipeline: preprocessing, GAN-enhanced class balancing, forecasting, SHAP and counterfactual explanation, personal exposure, health-risk assessment, cumulative exposure and alerts via a FastAPI backend and React dashboard. The forecasting and explainability components of the pipeline have been validated on held-out data; the personal exposure, health risk and cumulative exposure components are implemented and internally consistent, but as rule-based systems requiring further outcome validation. the immediate priorities hence lie in time-aware backtesting of the forecasting module, replacing the simulated exposure trajectory with a true multi-day forecast, adding live sensor/IoT and wearables tracking for personal exposure assessment, validation of health

risk weights against an independent clinical or epidemiological standard, large-scale deployment across the five Indian cities as a proof-of-concept, latency/usability testing of the deployed system, user-centered evaluations, and the addition of privacy-preserving personalization — transitioning the system from a single-region demo to a truly global, forecast-enabled, explainable and personalized environmental health platform.

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